Project Report

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Project Title:	House Price Prediction using PyTorch
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1 Project Overview

The goal of this project is to build a ML model which predicts house prices in USD, based on various characteristics of a property. Thus, an imported dataset is used, which contains house features and the corresponding sale prices. The house attributes used for training are analyzed further onto the next unit.

In order to evaluate the ML model, several visualisation metrics are provided, for instance a scatter plot of the actual versus the predicted prices, a price matrix showing the actual versus the predicted price ranges and two confusion matrices, after grouping prices into categories(Low/Medium/High).

The dataset used for training originates from Kaggle and more specifically it can be found in this link. It contains information about residential homes in Ames, Iowa, USA. The dataset includes 79 explanatory variables describing different aspects of each house. Each row represents one house sale.

As for the programming language, **Python** was preferred. Alongside Python, some libraries were also included. These were *numpy* for the numerical operations of matrices, *matplotlib* for the visualisation of the results, *pandas* for the dataset handling, *PyTorch* and *scikit-learn* for the ML model implementation and the confusion matrix creation.

The whole script is written in a jupyter notebook, along with a few explanations and comments. The screenshots of the results used in the report can also be found in the notebook. The project is uploaded as well on Github here. For the setup instructions, please refer to the README file.

2 Dataset Details

2.1 House Attributes

The dataset includes many categories of information about each house.

1. Property size: size of the lot in square feet, above-ground living area and total basememt size
2. House structure: year it was built, total condition, material quality, year of last remodeling, number of rooms
3. Rooms and layout: number of bedrooms, kitchens, bathrooms
4. Garage information: capacity, type
5. Exterior and Construction: building materials and architecture style
6. Location: neighborhood and proximity to main roads and railroads.

The dataset includes information about 1460 houses and their 79 features. There is only one target variable which is called Sale Price. The data is either numerical or text labels, which must be converted into numbers using one-hot encoding before training.

2.2 Imported Files

The downloaded folder from Kaggle includes several files needed for the model training and testing:

- The **train.csv** file contains the main dataset used for the model training. It is structured as a matrix with an ID for each house, certain values for the attributes analyzed above and its final sale price.

- The **test.csv** file used for testing the model predictions. It contains new house data, but without prices.

- The **sample_submission.csv** file which suggests the required format for prediction files using the **train.csv** data.

- The **data_description.txt** file explaining what each variable means and the correlation with a house feature.

3 Experimental Results

In order to assess the model's performance, the **train.csv** dataset is divided into training and validation set, because it contains the real housing prices which can then be compared with the predicted ones.

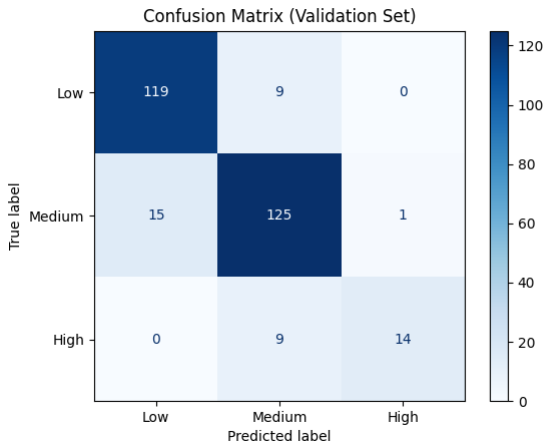

Figure 1: Confusion Matrix

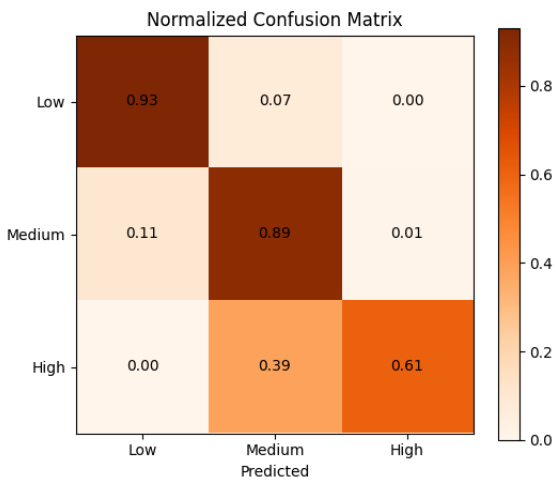

Figure 2: Normalized Confusion Matrix

By observing the confusion matrices, it is deducted that the model's accuracy is generally high, especially for Low and Medium price houses. However, it sometimes underestimates expensive houses.
Numerically, the model's accuracy is 93% for low price properties, 89% for medium price and 61% for high price ones. The rest 39% of wrong predicted expensive houses are misclassified as Medium price.

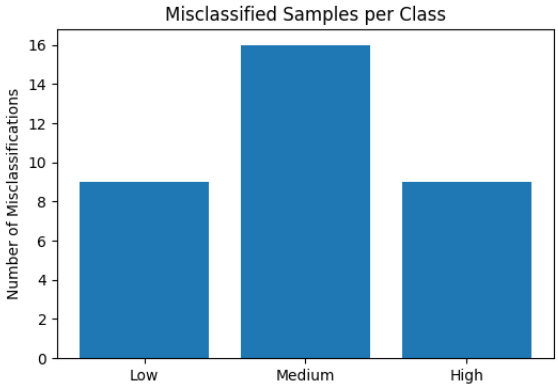

Figure 3: Misclassified samples per Class

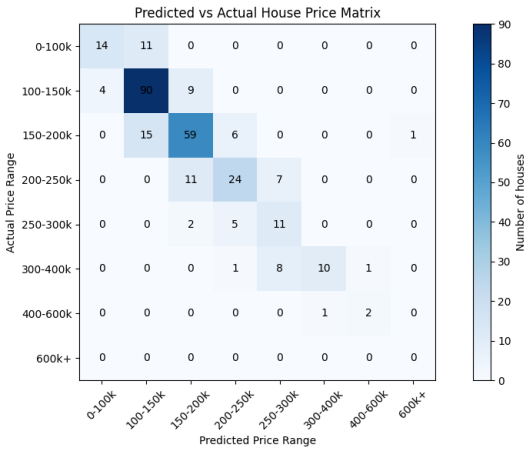

Figure 4: Predicted vs Actual House Price Matrix

The model predicts almost always correctly the houses that cost between 100-150 thousands USD and quite often those between 150-200 thousands USD. If a price is incorrectly estimated, the fault is mostly 50-100 thousands USD more or less. The price difference is rarely more significant. The percentage of misclassification is noticeable at around 250-300 thousands USD, where 11 houses are priced within their exact price range, but 15 houses in total are estimated to belong to a slightly different price range.

Figure 5: Predicted vs Actual Sale Prices Diagram

As discussed previously, most houses at around 100-250 thousands USD are predicted quite accurately. For

properties above 300 thousands USD, predictions become more scattered. Nevertheless, the extreme cases are rare, with the most noticeable one a 150 thousands USD house to be wrongly estimated as near 1 million.

```
Classification Report:
               precision    recall  f1-score   support

          Low       0.89      0.93      0.91       128
       Medium       0.87      0.89      0.88       141
         High       0.93      0.61      0.74        23

     accuracy                           0.88       292
    macro avg       0.90      0.81      0.84       292
 weighted avg       0.88      0.88      0.88       292
```

Figure 6: Classification Report

The overall accuracy, given that there are 292 validation samples is calculated:
$119 + 125 + 14 = 258$ correct assumptions, result which is equal to approximately **88%** accuracy.

However, supposed the model is working sufficiently good, the sale prices can be generated for the **test.csv** file and they can be saved into another file named **submission.csv**. The results of this calculation are shared in the jupyter notebook.

4 Conclusion

This project aimed to build a ML model which is trained to calculate a house sale price and later the model was applied for such a purpose. The Neural Network was implemented mainly by using the PyTorch library, while the whole script is written in Python. The total accuracy achieved was approximately **88%**.
The model predicts low and medium house prices with high accuracy, achieving around **93%** and **89%** correct classifications respectively. However, the performance decreases for high-priced houses, where only **61%** are correctly classified. The confusion matrix and scatter plot show that the model tends to underestimate expensive houses, often predicting them as medium-priced properties. This behavior is likely caused by data imbalance and the skewed distribution of house prices in the dataset.
Given the model created, the prices of the dataset **test.csv** were calculated and they can be compared with the sample file included inside the downloaded folder from Kaggle.
In the future, there might be added some techniques that let the model correctly process the bias in the imported data.